

Figure 8d. Summary of Hypothesis Tests

Hypothesis	Statement	Statistical Test	p-value	Sig.	Decision
H□a	LLaMA-4 Scout 17B > LLaMA-3.1 8B on Mathematics	Mann-Whitney U (one-tailed)	0.0006	***	Supported
H□b	LLaMA-4 Scout 17B > LLaMA-3.1 8B on Factual	Mann-Whitney U (one-tailed)	0.0130	*	Supported
H□c	LLaMA-4 Scout 17B > LLaMA-3.1 8B on Commonsense	Mann-Whitney U (one-tailed)	0.0159	*	Supported
H□	Overall faithfulness distributions differ by model	Pearson χ^2 (two-tailed)	2.82e-05	***	Supported
H□	Answer change rate differs significantly by model	Pearson χ^2 (two-tailed)	2.82e-05	***	Supported

Significance codes: *** $p < .001$ ** $p < .01$ * $p < .05$ ns $p \geq .05$ | $\alpha = .05$